

EE 595H : Stochastic Models

- A. This is **NOT** an open book exam. The exam is for **25 Marks** and of duration **75 Minutes**. The marks for each question is indicated at the beginning of the question.
- B. The question paper consists of **THREE** questions printed on **ONE** A4 side. Answer to each question should begin on a **FRESH** side of the answer book.

Question 1. (8 marks)

Consider the Markov chain with states $S = \{1, 2, 3\}$ that has the following transition matrix:

$$P = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

1. Draw the state transition diagram for this Markov chain.
2. Given $P(X_0 = 1) = P(X_0 = 2) = \frac{1}{4}$, find $P(X_0 = 3, X_1 = 2, X_2 = 3)$ and $P(X_1 = 2, X_2 = 3)$.
3. How many times, on the average, does the chain visit state 2 between two consecutive visits to state 1?

Question 2. (7 marks)

Let $Y_0, Y_1, Y_2, \dots$ be a sequence of independent die-rolls (of a six sided dice). Let $X_0, X_1, X_2, \dots$ be the stochastic process defined as $X_n = \max(Y_0, Y_1, \dots, Y_n)$.

- (a) Explain why $X_0, X_1, X_2, \dots$ is a Markov chain.
- (b) Compute the transition matrix P .
- (c) Find the initial distribution $\mu^{(0)}$.
- (d) Supposing that $X_0 = 3$, find the expected time until 6 is reached?

Question 3. (10 marks)

You are given a uniform random source producing independent uniform random numbers $U_0, U_1, U_2, \dots$

1. Find a method for simulating the random variable Y with density $f_Y(t) = e^{-\frac{|t|}{2}}$.
2. Describe how to use the rejection method with samples of Y to simulate a random variable X with the standard normal density (zero mean, unit variance).
3. What is the average number of source random numbers consumed per sample of X .